Experiment No. 7

Continuous-Time Fourier Transform and Spectrum

OBJECTIVE:

- To determine the magnitude and phase spectra of continuous-time signals and analyze their significant spectral components.
- To verify important Fourier transform properties and examine their effects on the corresponding time-domain and frequency-domain representations.

I. To determine the magnitude and phase spectra of continuous-time signals and analyze their significant spectral components.

Experimental Procedure:

- Define the continuous-time simulation vector and sampling interval.
- Generate the three signals.
- Compute the FFT and apply fftshift.
- Scale the transform using the sampling interval.
- Plot the time-domain waveform and magnitude spectrum for each signal.

MATLAB Coding:

```
clc; clear; close all;
%% 7(ii) Magnitude and Phase Spectrum

t = -2:0.001:2;
dt = t(2)-t(1);
x = exp(-t.^2).*cos(2*pi*3*t);

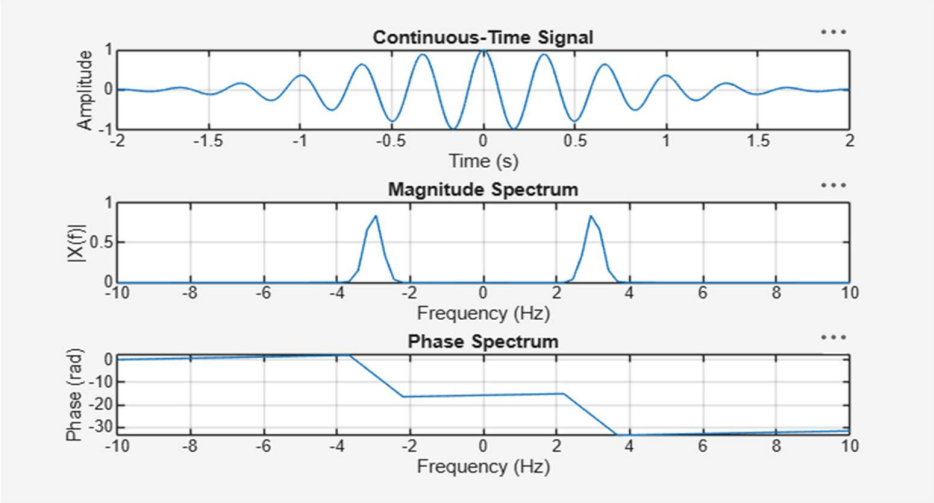
N = 4096;
X = fftshift(fft(x,N))*dt;
f = (-N/2:N/2-1)/(N*dt);

magX = abs(X);
phaseX = unwrap(angle(X));

figure;
subplot(3,1,1);
plot(t,x); grid on;
xlabel('Time (s)'); ylabel('Amplitude');
title('Continuous-Time Signal');

subplot(3,1,2);
plot(f,magX); xlim([-10 10]); grid on;
xlabel('Frequency (Hz)'); ylabel('|X(f)|');
title('Magnitude Spectrum');
```

Results:



II. Analysis of Signal Bandwidth and Spectral Components for Communication Applications

Experimental Procedure:

- Set fs = 2000 Hz and generate the composite signal.
- Compute the FFT and shift the spectrum.
- Plot the magnitude spectrum from −400 to 400 Hz.
- Apply the 5% magnitude threshold.
- Determine the approximate one-sided occupied bandwidth.

MATLAB Coding:

B Drive/exp_7c.m

```
clc; clear; close all;
%% 7(iii) Verification of Fourier Transform Properties

t = -2:0.001:2;
dt = t(2)-t(1);
x = exp(-t.^2);

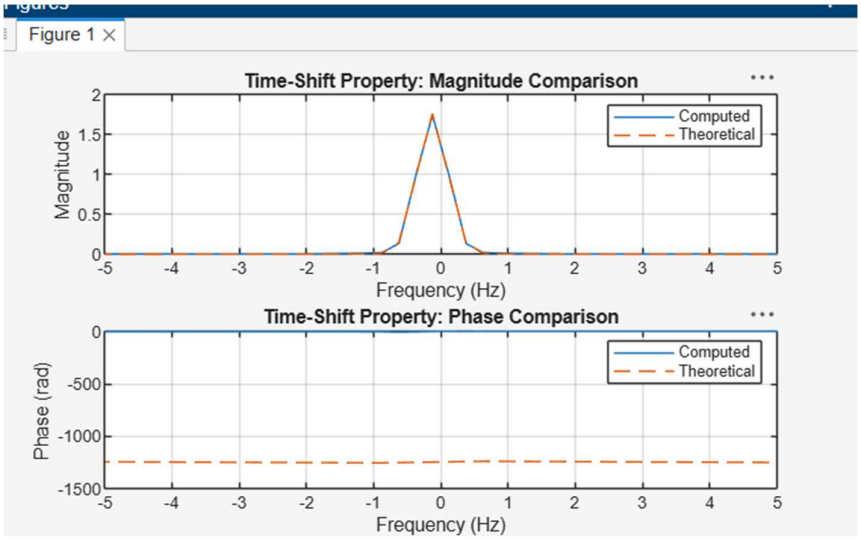
x_shift = interp1(t,x,t-0.4,'linear',0);

X = fftshift(fft(x))*dt;
N = length(x);
f = (-N/2:N/2-1)/(N*dt);

% Time-shift property
X_shift_theory = X.*exp(-1j*2*pi*f*0.4);
X_shift = fftshift(fft(x_shift))*dt;

subplot(2,1,1);
plot(f,abs(X_shift),f,abs(X_shift_theory),'--');
xlim([-5 5]); grid on;
xlabel('Frequency (Hz)'); ylabel('Magnitude');
title('Time-Shift Property: Magnitude Comparison');
legend('Computed','Theoretical');
```

Results:



Conclusion:

The continuous-time Fourier transform of representative signals was successfully obtained using FFT-based approximation, and their magnitude and phase spectra were analyzed to identify significant frequency components. The time-shift property was verified by comparing the computed and theoretical Fourier transforms, confirming that time shifting preserves magnitude while introducing a frequency-dependent phase shift. The spectral components and occupied bandwidth of a composite communication signal were also determined, demonstrating the importance of frequency-domain analysis in communication-system design.